

Embedded Linux & System Software Interview Handbook

Phase 5 – Interrupts, Timers & Deferred Work

This phase explains how the Linux kernel responds to hardware events, manages timers, and defers work for efficient execution. These concepts are fundamental for Linux device drivers and BSP development.

Chapter 12 – Interrupts, Timers & Deferred Work

Topics Covered

- Interrupt Fundamentals
- Interrupt Lifecycle
- Interrupt Descriptor Table (IDT)
- Interrupt Controllers (APIC, GIC)
- Edge vs Level Triggered Interrupts
- IRQ and IRQ Domains
- Interrupt Registration (request_irq)
- Interrupt Service Routine (ISR)
- Shared Interrupts
- Top Half vs Bottom Half
- SoftIRQ
- Tasklets
- Workqueues
- Threaded Interrupts
- Interrupt Affinity
- MSI and MSI-X
- High Resolution Timers (hrtimer)
- Kernel Timers
- Delayed Work
- Jiffies and Timekeeping
- Clock Sources
- Watchdog Timers
- Latency Optimization
- Interrupt Storms
- Interrupt Debugging
- /proc/interrupts
- ftrace and perf for IRQ Analysis

- Real Driver Examples (UART, Network, GPIO)
- Senior Interview Questions

Learning Outcomes

After completing this phase you will understand how hardware interrupts propagate through the Linux kernel, how deferred work mechanisms improve responsiveness, and how to debug interrupt latency and timing issues in production systems.

Phase	Coverage
Phase 1	Core Linux Kernel Fundamentals
Phase 2	Memory & Process Management
Phase 3	Filesystems & Storage
Phase 4	Linux Networking
Phase 5	Interrupts, Timers & Deferred Work
Estimated Size	200–300 pages when fully expanded
Target	Senior Embedded Linux / Kernel / BSP Interviews